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Modeling financial literacy through explainable machine learning and behavioral segmentation in emerging economies

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A R T I C L E I N F O

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A B S T R A C T

Digital financial services are changing how people behave financially but understanding how individuals use financial technology remains limited in emerging economies. This study investigates financial literacy and technology-driven financial behavior using machine learning analysis of 1067 adults in Bangladesh. Traditional demographic methods fail to capture the complexity of digital financial behavior, which hampers the design of effective interventions. By applying Random Forest, XGBoost, and k-means clustering validated with a silhouette score of 0.42, Davies-Bouldin Index of 1.08, and Calinski-Harabasz Index of 287.3, we achieved moderate predictive performance with an F1-score of 0.52, a 58 % improvement over random guessing, and gained important insights into technology-influenced financial behavior. SHAP analysis identified institutional trust, digital comfort, and income as key predictors, with trust showing an importance value of 0.18 compared to education at 0.09, challenging typical demographic assumptions. Three distinct behavioral groups emerged: Digitally Literate Planners (34 % of the sample, average financial knowledge 7.8/10), Informally Active but Underskilled (41 %, knowledge 5.4/10), and Digitally Excluded Traditionalists (25 %, knowledge 3.1/10). Results show that rural users had 7.3 % higher financial literacy than urban users, while formal education did not significantly impact literacy levels. Weak correlations between financial knowledge and actual behaviors, with coefficients below 0.10, suggest that understanding alone does not lead to effective financial practices. The findings provide frameworks for tailored interventions based on behavioral types instead of stereotypes, including resource allocation of 60 % for support, 30 % for transitional programs, and 10 % for advanced services, to improve intervention effectiveness for diverse users in developing economies.

1. Introduction

Financial literacy is crucial for economic well-being worldwide, but traditional assessment methods often overlook many of its aspects and fail to provide valuable insights for targeted actions (Lusardi & Messy, 2023; Lusardi & Streeter, 2023). People with higher financial literacy tend to achieve better financial results, including saving more, managing debt more effectively, preparing better for retirement, and being more resilient to economic shocks. As financial products become more complex and innovative, the ability to understand financial ideas, compare options, and navigate systems has become crucial for economic participation and success. The importance of financial literacy is especially clear in developing countries, where rapid digital changes are expanding access to financial services but also creating new challenges for people with limited experience in formal financial systems (Choung et al., 2023; Clark et al., 2025). Mobile banking, digital payments, and fintech are transforming financial environments in emerging markets; however, these technological advances do not automatically improve financial skills unless they are backed by basic understanding, proper institutional support, and user trust in digital platforms (Widyastuti

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et al., 2024; Adel, 2024). The digital financial gap is not just about access to technology, but also about fundamental gaps in knowledge, skills, trust, and confidence needed to navigate increasingly complex financial systems safely and effectively (Koskelainen et al., 2023). In Bangladesh, where mobile financial services are rapidly growing alongside ongoing financial exclusion, understanding the complex links between demographic traits, digital access, and financial literacy has become crucial for shaping policy and designing effective interventions (Hossain et al., 2020; Khalily, 2008). Despite notable progress in adopting mobile financial services, significant gaps remain in individuals' ability to use digital financial tools effectively and securely. This issue is especially troubling because simply having access to digital platforms does not guarantee proper usage without basic knowledge of financial concepts and trust in institutions (Choung et al., 2023). The lack of financial education is even more evident in rural and semi-urban areas, where infrastructural challenges and informational gaps continue to hinder financial empowerment. Addressing these issues requires innovative approaches to assess financial literacy and create interventions that consider Bangladesh's unique socioeconomic and technological environment (Arora & Sarker, 2025; Khalily, 2008; Rabeta & Sumi, 2023).

Traditional financial literacy research in developing countries has mainly depended on descriptive surveys and simple regression models, focusing mostly on demographic factors like age, education, and income (Zaimovic et al., 2023). These traditional methods, while offering useful initial insights, have clear limitations in capturing the complex, non-linear, and multidimensional relationships that shape financial behavior in fast-changing digital environments (Singh et al., 2020). Recent research has started using machine learning (ML) models—including Random Forest, XGBoost, and LightGBM—to improve prediction accuracy, classification, and understanding of important features in complex socioeconomic data (Garson, 2021; Imani et al., 2025). These advanced methods provide distinct benefits that suit financial literacy studies in diverse populations. ML algorithms are especially good at identifying non-linear relationships and threshold effects in financial behavior, automatically detecting complex interactions between variables without needing to specify every possible relationship beforehand, accommodating population differences through natural behavioral segmentation, and offering clear insights through explainable AI tools like SHAP values that policymakers and practitioners can easily understand and use (Jarupunphol et al., 2024; Yang & Xie, 2025). Research shows that combining clustering methods with supervised ML can greatly improve prediction accuracy in social science studies, and ensemble techniques like XGBoost tend to outperform traditional models in predicting human capital readiness in rural areas (Jarupunphol et al., 2024; Yang & Xie, 2025).

Theoretical frameworks surrounding financial literacy have evolved considerably from simple knowledge-based ideas to include behavioral insights and digital aspects. Instead of viewing literacy as just a binary or scalar variable, modern scholars see it as a multidimensional concept that involves cognitive understanding, behavioral actions, emotional attitudes, and the ability to adapt to different contexts (Bayakhmetova et al., 2025; Koskelainen et al., 2023). This view is based on behavioral finance theory, which highlights that financial behavior is influenced not only by rational calculation but also by cognitive biases, heuristics, and emotional factors that systematically affect choices even among knowledgeable people (Chawla et al., 2020; Singh et al., 2020). The term "financial aliteracy" refers to situations where individuals greatly overestimate their financial skills, which can lead to risky or irrational financial decisions. This issue is especially important in digital settings where superficial access to tools like mobile banking apps does not always mean proficient use (Chawla et al., 2020; Clark et al., 2025). Research shows that financial behavior mainly mediates the link between financial knowledge and perceived wellbeing, highlighting the need for behaviorally based measurement frameworks that go beyond just testing knowledge (Sabri et al., 2022). The financial capability approach expands this view by defining financial wellbeing as requiring both ability—knowledge and skills—and opportunity—access to suitable financial services and institutional support—to reach desired financial results (Koskelainen et al., 2023; Zaimovic et al., 2023).

Comparative research across various cultural settings consistently indicates that digital financial literacy is often a more reliable predictor of sound financial behavior than traditional literacy measures alone (Choung et al., 2023; Widyastuti et al., 2024). In Bangladesh, where women's access to financial services is frequently mediated through family or spousal networks, understanding these dynamics is essential for creating truly inclusive interventions (Khalily, 2008). The sociocultural and gendered aspects of financial literacy have gained increasing attention, with studies documenting ongoing gender gaps driven by complex factors such as social norms, confidence differences, and disparities in educational exposure (Haag & Brahm, 2025; Roshid & Le Ha, 2024). Evidence from around the world consistently shows a strong positive link between financial literacy and better financial behaviors, with those possessing higher financial knowledge more likely to engage in long-term planning, responsible borrowing, and regular savings (Lusardi & Messy, 2023; Rabeta & Sumi, 2023; Sabri et al., 2022). Recent studies also reveal that both digital and traditional financial literacy positively impact financial well-being in emerging markets, emphasizing the need for comprehensive literacy assessments that consider multiple aspects of financial capability (Kamble et al., 2024). Together, behavioral finance theory and the financial capability approach provide the conceptual foundation for this study's clustering and predictive modeling strategy.

Despite expanding access to digital financial tools across Bangladesh and increasing recognition of financial literacy's importance, significant gaps remain in both understanding and methodological approaches. Financial literacy is still underexplored among socioeconomically vulnerable groups in Bangladesh and similar developing countries, especially affecting low-income communities, rural residents, and undereducated populations (Hossain et al., 2020; Kamble et al., 2024; Rabeta & Sumi, 2023). Most notably, few studies have used ML approaches specifically designed to address socio-demographic differences within Bangladesh's developing economy context. Moreover, existing research rarely employs advanced analytics to systematically map financial literacy across behavioral groups or to offer personalized insights for targeted intervention strategies. The literature does not thoroughly investigate gendered and rural-urban digital divides using predictive analytics, which could help inform targeted interventions instead of generic population-level programs. Additionally, most studies do not apply explainable AI techniques that translate complex algorithmic results into transparent, policy-relevant insights that practitioners and policymakers can effectively use in resource-limited settings. This highlights a significant gap, given the potential for AI-powered frameworks to model financial literacy not just as an outcome variable but as a complex interplay of socio-economic, behavioral, and digital factors.

This study addresses these gaps by developing and evaluating a ML framework to predict financial literacy levels among Bangladeshi adults. It examines how socioeconomic, demographic, and behavioral factors influence financial capability within the context of an emerging economy characterized by diverse educational backgrounds, varying income levels, and differential access to digital financial services. The research integrates behavioral clustering with supervised ML to identify natural population segments while ensuring interpretability through SHAP analysis. This approach represents an innovative methodological combination of unsupervised and supervised learning techniques within the context of financial literacy in Bangladesh. By analyzing gendered patterns, rural-urban differences, and educational effects using comprehensive primary data, the study offers actionable frameworks for targeted interventions based on behavioral typologies rather than demographic stereotypes. The theoretical contribution conceptualizes financial literacy as a dynamic, multidimensional construct shaped by digital behavior patterns, contextual factors, and institutional trust. This work advances perspectives in behavioral finance that emphasize the knowledge-behavior gap in financial decision-making (Koskelainen et al., 2023; Zaimovic et al., 2023).

The primary aim of this study is to develop and systematically evaluate a predictive ML framework designed to assess financial literacy levels among adults in Bangladesh. The specific objectives encompass: identifying key socio-demographic and digital access variables that influence financial literacy within the Bangladeshi context through feature importance analysis and explainable AI techniques; implementing advanced ML models—namely, Random Forest, XGBoost, and Decision Tree—to classify financial literacy with superior predictive accuracy and interpretability in comparison to conventional regression methods; analyzing behavioral clusters via unsupervised learning to identify distinct population segments that support targeted financial education initiatives beyond traditional demographic classifications; and investigating the correlations between financial knowledge and actual financial behaviors to enhance both theoretical understanding and practical intervention strategies. This research makes methodological contributions by incorporating explainable AI models in a domain traditionally dominated by static regression analyses and descriptive survey summaries (Garson, 2021); conceptual contributions by framing financial literacy as a complex function of interconnected variables influenced by digital transformation and institutional contexts (Koskelainen et al., 2023; Zaimovic et al., 2023); and practical contributions by offering actionable insights for financial educators, policymakers, and service providers aiming to optimize educational content and intervention strategies for Bangladesh's diverse population segments characterized by varying levels of formal education, digital access, and engagement with formal financial institutions.

2. Materials and methods

2.1. Study design and sampling

This research employed a cross-sectional, quantitative design combining primary data collection through structured surveys with ML-based predictive modeling to evaluate financial literacy among adults in Bangladesh. The study was conducted from March to June 2024 and targeted adults aged 18–60 with access to digital platforms across diverse geographic and socioeconomic backgrounds. This approach improves on traditional correlation-based financial literacy studies by integrating robust survey methods with computational analytics to uncover complex, nonlinear relationships between socioeconomic factors and financial outcomes that standard regression models often overlook (Imani et al., 2025; Singh et al., 2020). The framework followed a step-by-step process involving primary data collection via online surveys, thorough data preprocessing and feature engineering, development and assessment of supervised ML models, and unsupervised behavioral clustering for population segmentation. This setup enables both predictive modeling and natural behavioral segmentation, which traditional statistical methods often miss, particularly in the diverse populations typical of developing economies (Yang & Xie, 2025). Fig. 1 presents the complete methodological workflow outlining key stages from study design to policy implications.

A stratified random sampling technique was used with proportional allocation based on Bangladesh's documented urban-rural population split, maintaining a 45:55 urban-rural ratio to ensure representation across key demographic groups. The target group included adults aged 18–60 with digital access, aligning with the focus on individuals who could benefit from existing and emerging digital financial services. Sample size was determined through statistical power analysis

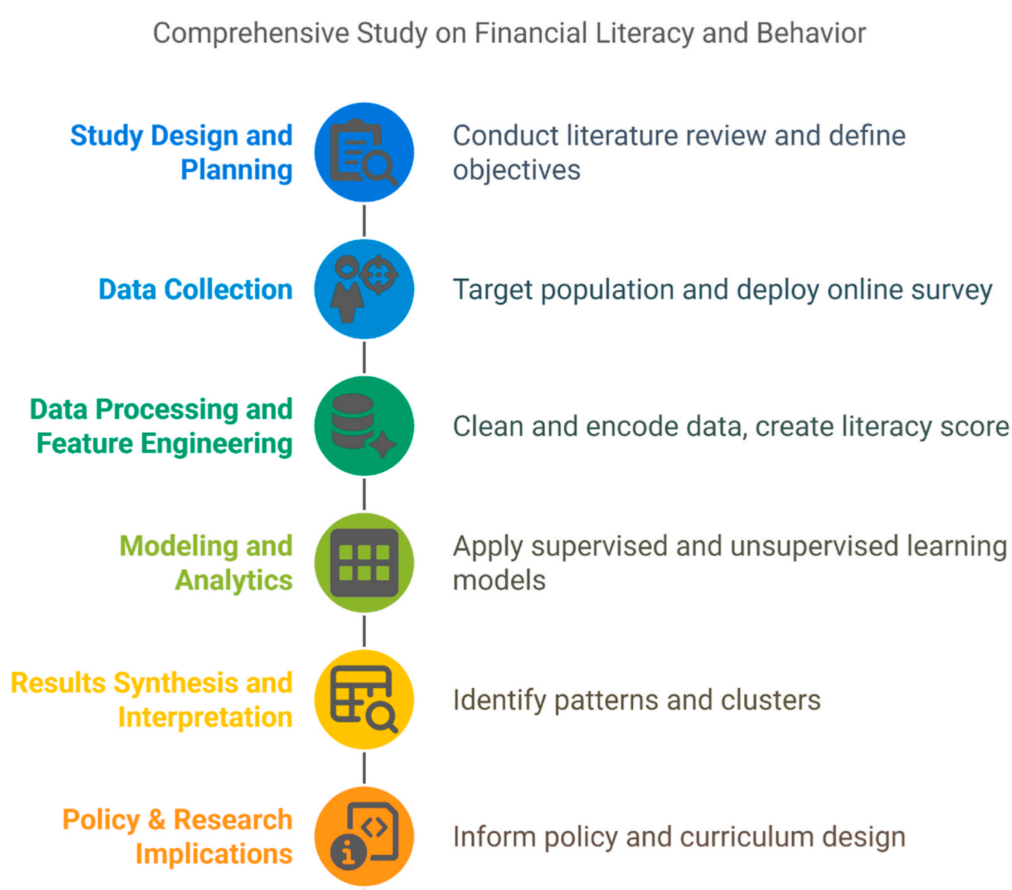

Fig. 1. Methodological workflow outlining key stages from study design to policy implications in the financial literacy analysis.

conducted beforehand using G*Power 3.1 software for logistic regression with three predictor categories. The parameters included an alpha significance level of 0.05, a power of 0.80, and a medium effect size corresponding to an odds ratio of 1.5. This analysis showed that at least 786 participants were needed to detect significant effects with enough statistical power. The actual sample of 1067 valid responses surpassed this minimum by 35.7 %, ensuring sufficient power for multivariate analysis. For ML purposes, the common guideline of at least ten observations per predictor variable was met given the 28 predictors used, resulting in about 38 observations per variable—well above the minimum—and providing enough data for training and testing models without overfitting concerns. The digital-only survey method was chosen for its practical advantages in reaching geographically dispersed populations, cost-effectiveness within the research budget, and its alignment with the study's focus on digitally accessible groups who are the primary users of mobile financial services. During the March–June 2024 data collection period, digital outreach was the most feasible method given the rising rates of mobile and internet use across Bangladesh. However, this approach has limitations, such as digital exclusion, since the most financially vulnerable individuals without reliable internet access might be underrepresented, potentially biasing results toward more digitally engaged groups.

2.2. Data collection and survey instrument

This study involved collecting original primary data rather than extracting it from existing databases or doing secondary analysis of previously gathered datasets. All data was directly gathered from Bangladeshi respondents using a custom-designed survey instrument administered online during the designated study period. Choosing to collect primary data was methodologically crucial to ensure the questions were suitable for Bangladesh's specific financial environment, including mobile money services like bKash and Nagad, informal savings groups, and rural cooperative banking practices. Data quality was preserved through multiple validation steps, such as response time monitoring with surveys completed in less than 8 min flagged for review, attention check questions embedded throughout the survey, logical consistency checks comparing related responses, and IP address verification to prevent duplicate submissions. The questionnaire was created specifically for this research, guided by the internationally recognized OECD/INFE financial literacy assessment framework, while extensively adapting content to match Bangladesh's unique financial context (OECD, 2022). The final survey included 30 items across six domains: demographics (6 items including age, gender, education, employment, income, location), digital access and usage (5 items such as smartphone ownership, internet connectivity, mobile banking apps, payment frequency, trust), financial knowledge (10 items on interest calculations, inflation, risk diversification, investment basics, insurance fundamentals), behavioral traits (4 items regarding savings habits, budgeting, borrowing, planning), attitudinal dimensions (3 items on decision confidence, institutional trust, perceived security), and resource access (2 items about financial education sources and advice availability). The instrument was thoroughly validated through internal reliability checks using Cronbach's alpha for all subdomains, with results showing excellent consistency above 0.82, construct validity confirmed through exploratory factor analysis with proper factor loadings, pilot testing with 30 participants representing diverse demographics, and expert review by financial literacy researchers and practitioners in Bangladesh (Garson, 2021; Koskelainen et al., 2023).

2.3. Data preprocessing and variable operationalization

Prior to model development, the collected dataset underwent comprehensive preprocessing procedures designed to ensure optimal data quality for ML applications. All categorical features were transformed using one-hot encoding, and numerical variables were normalized using Min-Max scaling (0–1 range) to ensure comparability across measurement units. Missing data were minimal across the dataset, affecting less than 1.2 % of total responses, and were addressed using median imputation for numerical variables and mode imputation for categorical variables. Feature engineering played a vital role in enhancing the predictive capacity of ML models by capturing intricate relationships that extend beyond simple variable effects. Several interaction terms were developed based on Bangladesh-specific financial behavior patterns: income × digital access (technology adoption across economic strata), education × age (generational differences in financial capability), gender × household decision-making roles (cultural influences on financial authority), and urban-rural residence × digital comfort (geographic variations in technology adoption). The dependent variable, financial literacy level, was operationalized through a composite scoring approach integrating multiple assessment dimensions (Lusardi & Streeter, 2023; Sabri et al., 2022). The composite score was derived from ten core assessment items reflecting three pivotal dimensions: objective financial knowledge, including interest calculation skills, inflation comprehension, and risk diversification concepts; applied financial behaviors, such as budgeting practices, saving habits, and borrowing decision-making processes; and financial decision-making capabilities, including investment choices and insurance utilization patterns. Individual item scores were weighted in accordance with established financial literacy assessment protocols based on OECD guidelines, with scores aggregated to produce comprehensive literacy scores ranging from 0 to 10 (OECD, 2022). These scores were subsequently classified into three ordinal categories—Low, Moderate, and High literacy—using tertile-based classification to ensure balanced class representation for ML applications, yielding approximately 33 % classified as Low literacy, 52 % as Moderate literacy, and 15 % as High literacy.

2.4. ML model development

Three supervised learning classifiers were strategically selected based on empirical benchmarking studies in financial behavior modeling: Random Forest, XGBoost, and Decision Tree. The criteria for model selection prioritized predictive accuracy and F1-score, which balances precision and recall, particularly important for imbalanced datasets. Model interpretability was also a key consideration, as it is crucial for policy-relevant applications (Garson, 2021; Kelly & Xiu, 2023). Random Forest was chosen for its robust handling of categorical variables, resistance to overfitting through ensemble averaging, and ability to provide reliable feature importance rankings (Suarez-Lledo and Alvarez-Galvez, 2019). XGBoost was included based on research demonstrating its superior performance in predicting human capital readiness in rural communities and other socioeconomic outcomes in developing countries (Jarupunphol et al., 2024). Decision Tree was incorporated to provide high interpretability through transparent decision rules that policymakers can easily understand and implement.

The dataset was systematically divided using stratified sampling into a training set comprising 70 % of the data, a validation set comprising 10 % of the data for hyperparameter tuning, and a testing set comprising 20 % of the data reserved for the final unbiased performance evaluation. Stratified sampling ensured that the class distribution was preserved across all three partitions. Hyperparameter optimization was conducted through a comprehensive grid search combined with 5-fold cross-validation to identify optimal configurations while preventing overfitting. For Random Forest, the grid search explored n_estimators values of 100, 200, and 300; max_depth values of 10, 20, 30, and None; and min_samples_split values of 2, 5, and 10. For XGBoost, explored parameters included learning_rate values of 0.01, 0.05, and 0.1; max_depth values of 3, 5, and 7; and n_estimators values of 100, 200, and 300. For Decision Tree, max_depths of 5, 10, 15, and 20 were evaluated, along with min_samples_split values of 2, 5, 10, and 20.

Class imbalance, particularly the underrepresentation of high-literacy individuals at 15 % of the sample, was systematically addressed through resampling techniques applied exclusively to the training data. Synthetic Minority Over-sampling Technique (SMOTE) was employed to generate synthetic examples of minority classes by interpolating between existing minority class observations in feature space (Imani et al., 2025). Additionally, Adaptive Synthetic (ADASYN) sampling was applied, which generates synthetic samples with density distribution according to learning difficulty. These techniques improved class balance in the training set to approximately 30-40-30 distribution across low, moderate, and high literacy classes, while maintaining the original test set distribution for realistic performance evaluation.

Feature importance analysis utilized both traditional Gini importance metrics provided by tree-based algorithms and SHAP (SHapley Additive exPlanations) values to provide complementary perspectives on feature contributions (Crompton and Burke, 2023). SHAP analysis, grounded in cooperative game theory, assigns each feature an importance value for individual predictions by calculating the marginal contribution of each feature across all possible feature combinations. SHAP visualizations included summary plots showing global feature importance rankings, dependence plots illustrating relationships between individual features and predictions, interaction plots revealing compound effects between variable pairs, and force plots explaining individual predictions. This explainability framework transformed abstract algorithmic outputs into concrete, actionable insights regarding determinants of financial literacy.

2.5. Behavioral clustering analysis

Unsupervised learning utilizing k-means clustering was employed to identify natural population segments based on behavioral patterns and financial characteristics rather than predefined demographic categories. The optimal number of clusters was determined through three complementary approaches. Firstly, the elbow method examined the within-cluster sum of squares (WCSS) across k values from 2 to 8, identifying the point where the marginal reduction in WCSS diminishes. Secondly, silhouette analysis evaluated cluster cohesion and separation by measuring the similarity of each observation to its own cluster compared to other clusters. Thirdly, the gap statistic compared within-cluster dispersion to the expected values under a null distribution of data with no inherent clustering structure. These three methods converged on k equals 3 as the optimal choice, supported by a silhouette score of 0.42, indicating reasonable cluster separation, where values above 0.4 suggest meaningful structure. Additionally, the Davies-Bouldin Index of 1.08 indicates good clustering quality, as lower values below 1.5 are deemed acceptable, while the Calinski-Harabasz Index of 287.3 signifies well-defined clusters, with higher values reflecting greater separation among them.

K-means implementation employed the k-means++ initialization algorithm to optimize centroid selection, which probabilistically chooses initial centroids that are distant from each other to enhance convergence speed and solution quality. The algorithm was configured with a maximum of 300 iterations, a convergence criterion of 1e-4 based on centroid movement threshold, and 10 random initializations with different starting points to ensure the identification of the global optimum. Hierarchical clustering with Ward linkage was conducted as a validation step, producing dendrograms that corroborated similar segment structures with three-cluster solutions demonstrating clear separation. Principal Component Analysis (PCA) was subsequently applied to the clustered data, reducing dimensionality while maintaining 85 % of the variance, thereby facilitating visualization of cluster separation in two-dimensional space and confirming distinct population segments.

2.6. Model evaluation and ethical considerations

The performance of the model was assessed using multiple metrics appropriate for multi-class classification problems. Overall accuracy determined the proportion of correct classifications across all classes. Precision, recall, and F1-score were computed for each class individually and subsequently macro-averaged across classes to assign equal significance to each category, irrespective of sample size. The F1-score signifies the harmonic mean of precision and recall, offering a singular metric that balances these concerns and proves especially valuable for datasets exhibiting class imbalance. The Area Under the Receiver Operating Characteristic Curve (AUC-ROC) was calculated employing a one-vs-rest approach suitable for multi-class problems, measuring the model's capability to distinguish between classes across various decision thresholds. Confusion matrices provided detailed insights into misclassification patterns, indicating which classes were most frequently confused. All aforementioned performance metrics were derived from the held-out test set, which was excluded from the model training and hyperparameter tuning processes, thereby ensuring an unbiased evaluation of the model's generalization capacity.

Prior to data collection, comprehensive ethical clearance was granted by the Bangladesh Army International University of Science and Technology Ethics Review Board under approval number 2024-0102. Participation was entirely voluntary, with no remuneration or incentives offered. Digital informed consent was obtained from all respondents at the survey's outset, including explicit explanations of the study's purpose, data utilization, confidentiality measures, and the right to withdraw. Data anonymization procedures were implemented to ensure that no personally identifiable information was retained within the analytical dataset. All survey responses were assigned anonymous identification codes, with any identifiable data promptly separated from response data and destroyed following data collection. Results were reported in aggregate form, with minimum cell sizes of ten observations to prevent individual identification. Data security was maintained through encrypted storage, with access limited solely to authorized members of the research team.

3. Results

The results explore financial literacy patterns, modeling results, and behavioral segments from 1067 responses in Bangladesh. They aim to map literacy across demographics, identify predictors via ML, classify behavioral groups, and examine links between demographics and literacy. The findings challenge assumptions and offer insights for targeted interventions. The sample was 53 % male and 47 % female, reflecting Bangladesh's demographics and cultural factors affecting participation.

3.1. Financial literacy distributions across demographic groups

The geographic analysis revealed counterintuitive patterns challenging urban-centric policy assumptions. Fig. 2 shows rural participants marginally surpassed urban residents in financial literacy (rural mean: 5.73; urban mean: 5.34). Although this difference was not statistically significant ($F(1, 1065) = 2.31$, $p = 0.13$, $\eta^2 = 0.002$), the consistent pattern suggests that informal financial knowledge transmission in rural communities—through cooperative savings groups, agricultural credit associations, and traditional financial networks—may be more effective than previously acknowledged. This finding indicates that policy interventions should recognize existing informal financial capabilities rather than presuming knowledge deficiencies.

Educational attainment analysis revealed patterns that complicate traditional human capital theories. Figs. 3 and 4 show financial literacy score distributions across educational levels. No significant differences emerged in average scores (primary: 5.48, $SD = 1.82$; secondary: 5.52, $SD = 1.76$; higher secondary: 5.61, $SD = 1.79$; graduate: 5.68, $SD = 1.73$; $F(3, 1063) = 0.74$, $p = 0.59$, $\eta^2 = 0.002$). However, higher educational attainment was associated with more consistent scores, with the interquartile range narrowing from 3.2 points (primary) to 2.1

Fig. 2. Financial literacy scores by region.

points (graduate education). This suggests education may influence consistency of financial knowledge application rather than overall capability, and that practical financial knowledge in Bangladesh may be acquired through multiple pathways, including family transmission, community networks, and workplace experience.

Gender-education interaction analysis revealed more nuanced patterns that inform targeted intervention design and highlight the complexity of demographic influences on financial literacy outcomes. Fig. 5 presents a rose diagram illustrating these interaction effects, demonstrating that among female respondents, financial literacy scores peaked at the graduate education level with a mean of 6.12 (SD = 1.54), compared to means of 5.23 (SD = 1.91) for primary education, 5.41 (SD = 1.82) for secondary education, and 5.68 (SD = 1.76) for higher secondary education. Two-way ANOVA confirmed a significant interaction effect for women ($F(3, 499) = 4.83$, $p = 0.003$, $\eta^2 = 0.028$), suggesting that women derive greater financial literacy benefits from tertiary education compared to lower educational levels. This pattern may reflect cultural factors that limit women's access to informal financial learning opportunities, making formal education a more critical pathway for female financial capability development. In contrast, male participants displayed relatively consistent literacy performance across all education levels, with means ranging from 5.62 (SD = 1.75) for primary education to 5.74 (SD = 1.71) for graduate education, showing no statistically significant variation ($F(3, 562) = 0.91$, $p = 0.44$, $\eta^2 = 0.005$). This pattern indicates that men may acquire financial knowledge through diverse pathways beyond formal education, possibly including workplace exposure, business activities, and social networks that provide financial learning opportunities regardless of educational credentials. The gender-education interactions revealed in Fig. 5 highlight the need for differentiated intervention approaches that account for varying pathways to financial literacy development.

3.2. Relationships between financial knowledge, behavior, and demographics

Correlation analysis between financial knowledge assessments and behavioral indicators revealed weak associations that challenge conventional assumptions about the relationship between knowledge and behavior in financial literacy research and program development. Fig. 6 shows a comprehensive correlation matrix indicating that the ten financial knowledge questions—covering topics like interest calculations, inflation understanding, risk diversification, investment basics, borrowing decisions, insurance concepts, budgeting principles, savings strategies, emergency planning, and retirement preparation—exhibited weak individual correlations with behavioral indicators such as actual savings practices, budgeting behaviors, and emergency fund management. The absolute values of the correlation coefficients ranged from 0.02 to 0.09, with none reaching statistical significance at the 0.05 level.

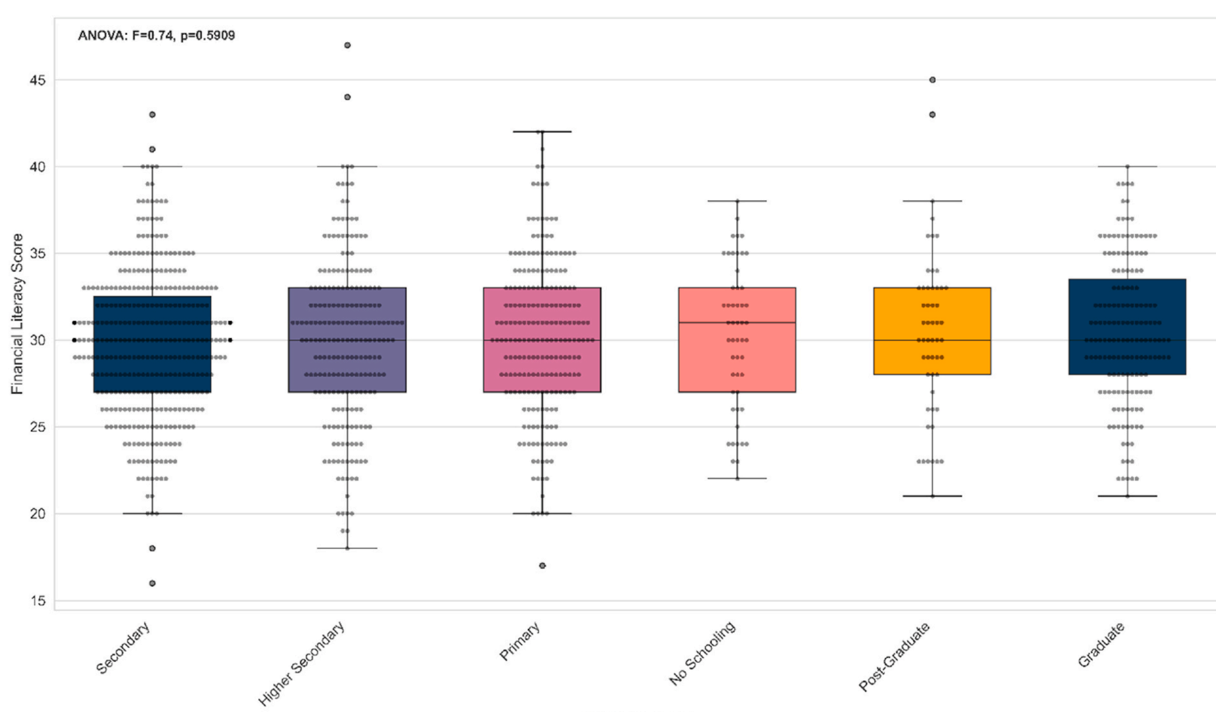

Fig. 3. Literacy distribution by education level with no significant differences.

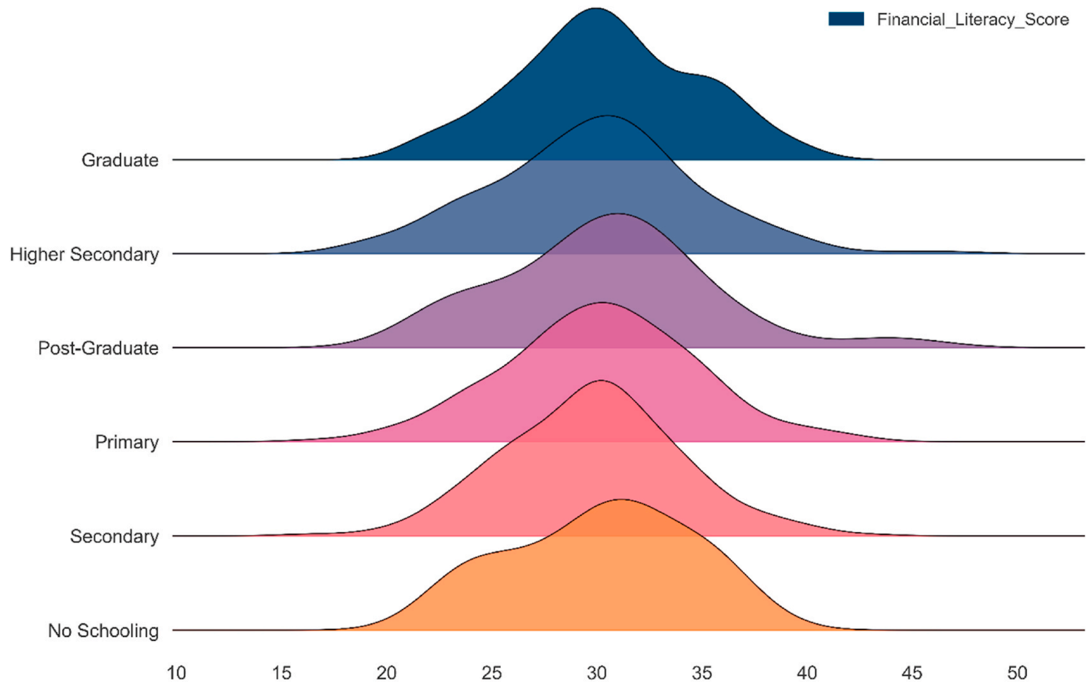

Fig. 4. Density of literacy scores across education groups.

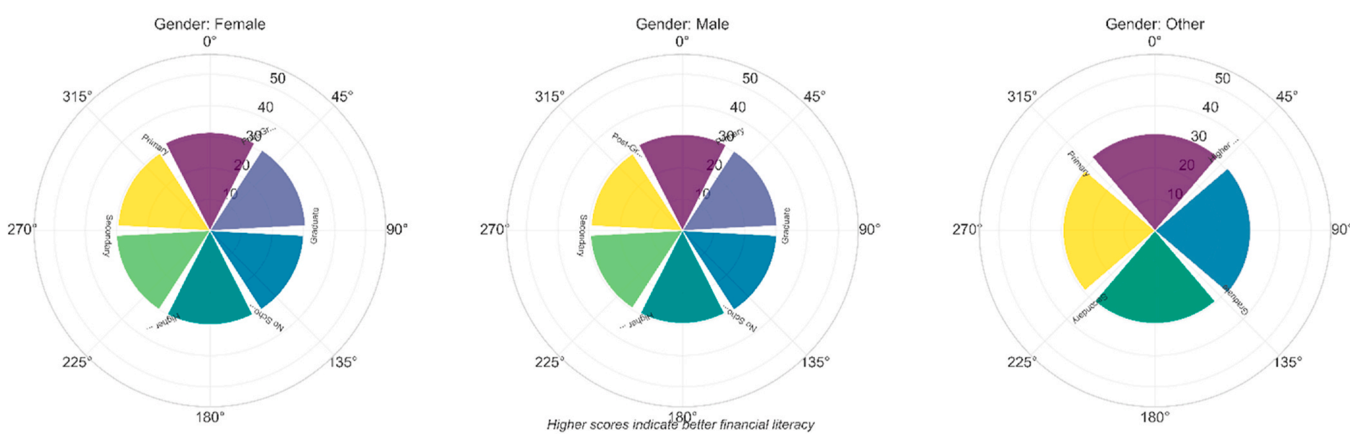

Fig. 5. Gender–education interaction effects on financial literacy.

This suggests that having theoretical financial knowledge does not automatically lead to practical financial actions, supporting views that highlight the importance of experiential learning, behavioral reinforcement, and contextual factors in financial education programs. The strongest relationships appeared between investment confidence measures and specific knowledge questions about risk assessment and diversification principles, with a correlation coefficient of 0.23 (95 % CI [0.17, 0.29], $p < 0.001$). This indicates partial alignment between understanding and confidence in specific financial areas. Still, these correlations are modest, implying that confidence in financial decision-making is influenced by factors beyond simply acquiring knowledge, such as prior experience with financial institutions, social support networks, cultural attitudes toward risk, and perceived access to reliable financial services.

Fig. 7 delineates the demographic similarity analysis, offering valuable insights into the relative significance of various personal characteristics in predicting financial literacy outcomes. The analysis reveals that education level and employment status exhibit the highest associations with financial literacy classification, each with similarity scores of approximately 36.86 %, thereby affirming their relevance. However, these relationships are moderate rather than deterministic. Age recorded a similarity score of 24.32 %, and marital status 18.47 %, suggesting moderate influence levels and emphasizing the importance of considering life stage contexts and family responsibilities within financial literacy initiatives, rather than relying solely on static demographic attributes. Gender demonstrated the lowest similarity score at 12.15 %, indicating minimal direct association with literacy outcomes when other factors are controlled. The analysis underscores that multiple factors collectively influence financial literacy outcomes, with no singular demographic characteristic emerging as a dependable predictor. Additionally, the combined effect of various demographic variables accounts for only a moderate proportion of variance in literacy classifications.

Fig. 8 depicts the intricate relationship between age and monthly income through a hexbin scatter plot that illustrates clustering patterns with significant implications for targeting financial inclusion initiatives. While the overall Pearson correlation between age and income was weak ($r = 0.01$, $p = 0.89$), the visualization exposes distinct clustering patterns among individuals aged 25–45 earning between 20,000–30,000 Bangladeshi Taka monthly, with this cluster comprising approximately 38 % of the total sample. These income-age clusters may represent

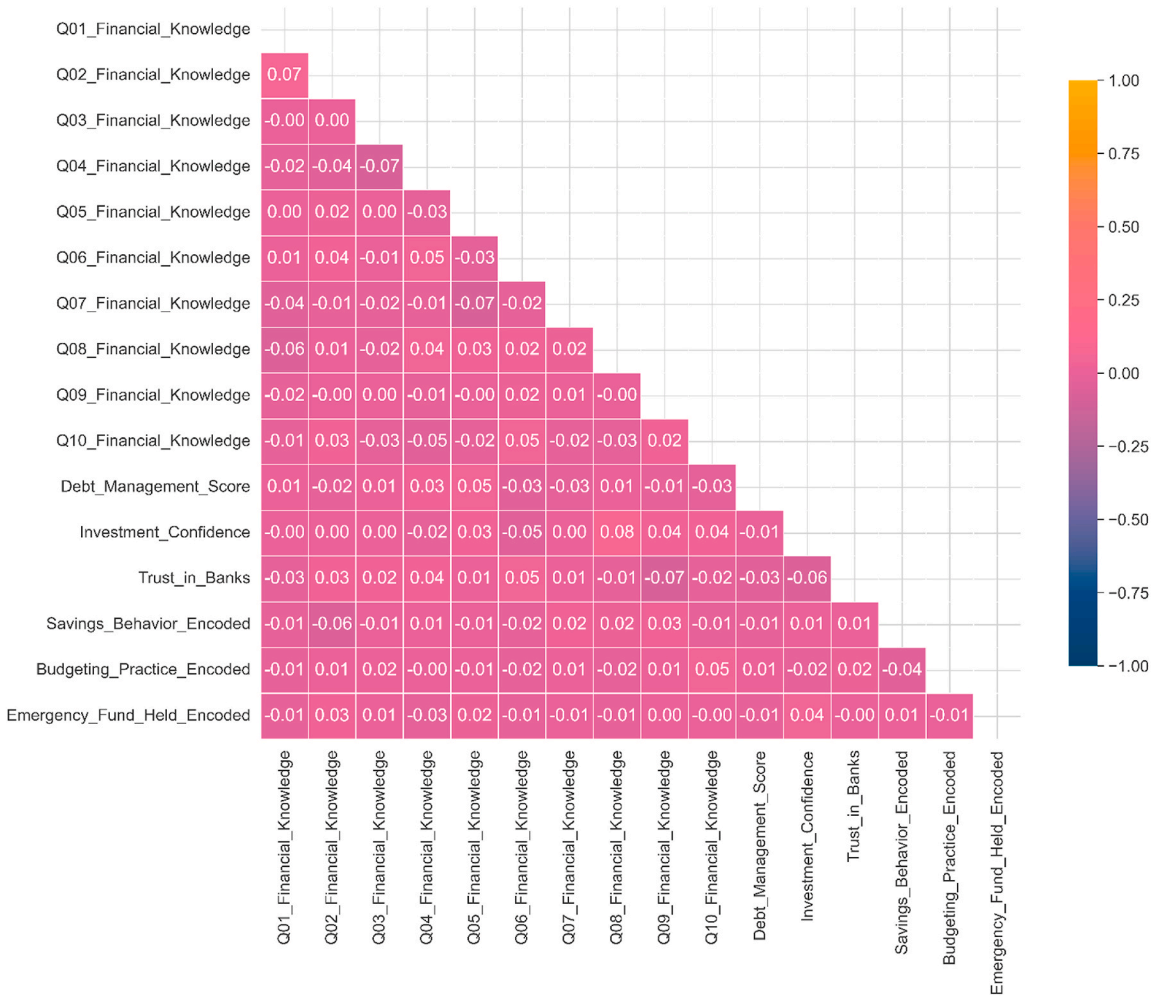

Fig. 6. Correlation matrix of knowledge and behavior indicators.

Bangladesh's emerging working-class segment, whose financial literacy development is likely influenced by factors such as access to digital technology, employment stability, exposure to formal financial services through workplace benefits, and integration within urban banking infrastructure. The clustering patterns indicate that age-income combinations may provide more insightful information for targeting purposes than either variable alone, thereby presenting opportunities for the development of segment-specific interventions that consider the joint distribution of these demographic attributes.

3.3. ML model performance and feature importance

Three supervised learning models—Random Forest, XGBoost, and Decision Tree—were systematically trained and evaluated to predict financial literacy classifications. Performance assessment was conducted on an independent holdout test set comprising 20 % of the sample to ensure an unbiased evaluation of their generalization capabilities. Fig. 9 shows comprehensive performance metrics across all three classifiers, with XGBoost achieving superior overall performance F1-score of 0.52 and an AUC-ROC of 0.527—marginally outperforming Random Forest, which had F1-score of 0.50 and an AUC-ROC of 0.514, and significantly outperforming Decision Tree, which had F1-score of 0.47 and an AUC-ROC of 0.496. Overall accuracy ranged from 51 % for Decision Tree to 54 % for XGBoost. Although these performance levels seem modest, context is important: XGBoost's F1-score represents a 58 % improvement over random classification baseline (0.33), class imbalance (High literacy: 15 %) poses inherent prediction challenges, and social science behavioral prediction typically yields F1-scores between 0.45 and 0.65. Beyond predictive accuracy, these models are valuable for identifying feature relationships and offering interpretable insights for policy development.

Fig. 10 shows confusion matrix analysis for the Random Forest model, highlighting challenges in classifying individuals across three literacy categories. The model achieved 53 % accuracy for Moderate literacy, likely due to larger sample size and clearer traits. Low literacy accuracy was 47 %, while High literacy was only 10 %. The matrix shows 62 % of actual High literacy individuals were misclassified as Moderate, indicating difficulty distinguishing them with current variables. Similarly, 38 % of Low literacy were misclassified as Moderate. These patterns suggest the difficulty in identifying high literacy may be due to its rarity, subtle differences requiring better measurement, or unmeasured variables. Improving methods could involve ensemble strategies, cost-sensitive learning, or feature engineering to better discriminate the High literacy category.

Fig. 11 presents SHAP analysis providing crucial insights into feature importance. Monthly income emerged as the strongest predictor (SHAP: 0.31), followed by trust in banking institutions (0.18), age (0.14), sources of financial education (0.12), and digital comfort (0.11). Education level showed limited importance (0.09), reaffirming that formal education plays a surprisingly limited predictive role, while gender exhibited minimal direct capacity (0.06). The prominence of institutional trust as the second-strongest predictor underscores the importance of cultivating confidence in formal financial systems, suggesting that enhancing banking accessibility and institutional transparency

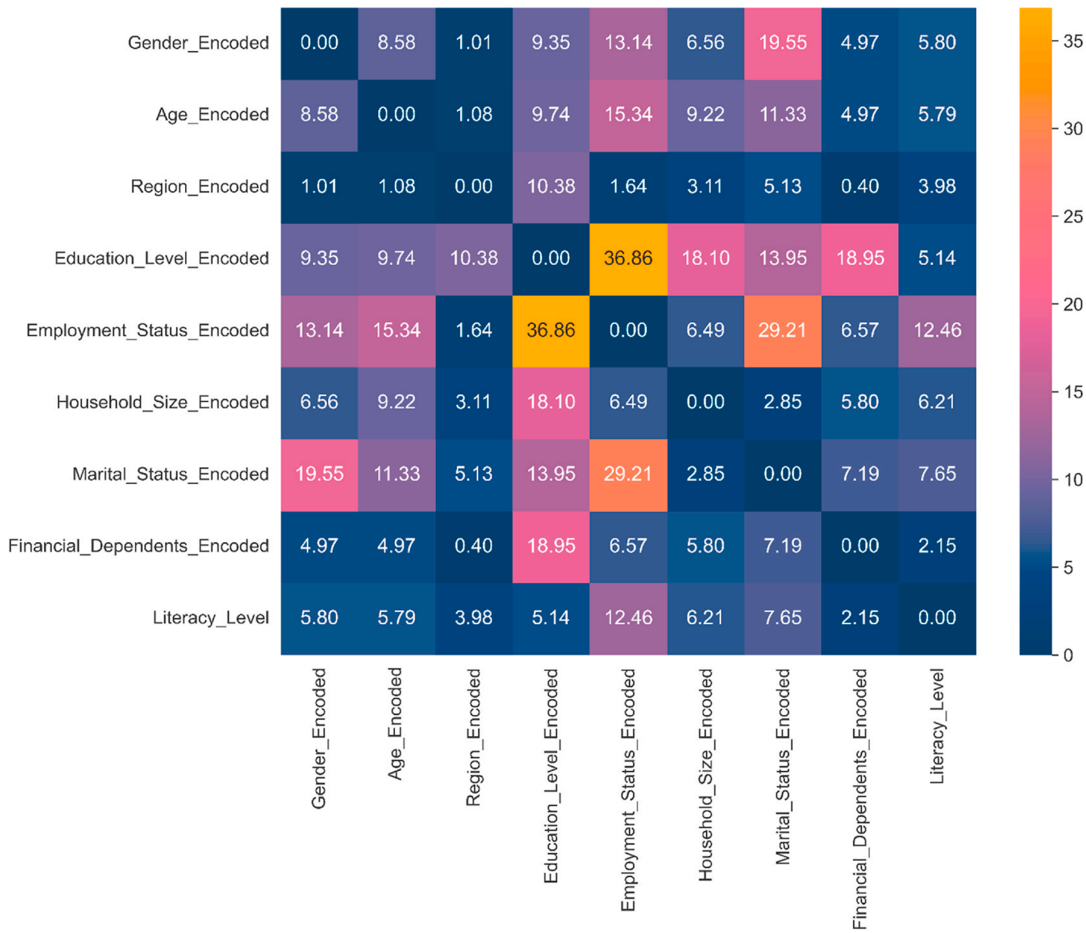

Fig. 7. Demographic similarity scores relative to literacy classification.

could substantially advance literacy outcomes.

Fig. 12 illustrates the interaction analysis among age, income, and predicted literacy through a three-dimensional visualization, which unveils complex relationships with significant implications for targeting strategies. The analysis indicates that individuals aged 35–50 with moderate income levels (20,000–30,000 BDT monthly) exhibit the highest predicted literacy probabilities, exceeding 0.65 for this demographic segment. Conversely, younger individuals under 30 with lower incomes below 15,000 BDT monthly demonstrate predicted literacy probabilities below 0.35, indicating higher risks of low literacy classification. Middle-aged individuals with very high incomes exceeding 40,000 BDT also display elevated literacy predictions, surpassing 0.60. This pattern likely reflects various underlying factors, including limited financial exposure among young adults who have had fewer opportunities to acquire practical experience, restricted access to digital financial services due to economic constraints that hinder meaningful engagement with formal financial systems, fewer opportunities for practical financial experience among younger populations who have yet to navigate major life financial decisions, and potential exclusion from formal financial systems that offer learning opportunities. The interactions between age and income suggest the necessity for targeted interventions that address both economic limitations and experiential learning opportunities for younger, economically disadvantaged demographic segments.

3.4. Behavioral segmentation and population clustering

Unsupervised learning techniques, including K-Means clustering, effectively identified three coherent behavioral segments within the population of Bangladesh, thereby offering actionable frameworks for the design of targeted interventions. The clustering analysis was validated through multiple metrics: a silhouette score of 0.42, indicating a reasonable degree of cluster separation, as values above 0.4 suggest meaningful structural delineation; a Davies-Bouldin Index of 1.08, indicating good clustering quality, with lower values representing better-defined clusters; and a Calinski-Harabasz Index of 287.3, demonstrating well-defined clusters, with higher values indicating greater separation between clusters relative to within-cluster dispersion. Fig. 13 presents comprehensive behavioral profiles across the three identified clusters, revealing distinct patterns in financial behaviors, digital engagement, and institutional relationships, rather than traditional demographic categories. Fig. 14 complements these profiles by comparing behavioral strengths across clusters, highlighting relative capabilities in savings, digital engagement, and formal banking utilization.

Cluster 1, called "Informally Active but Underskilled," makes up 41 % of participants (n = 437). It shows a mixed profile—strong in budgeting but weaker in emergency preparedness, formal financial services, and advanced planning. As shown in Fig. 14, this group has a 72 % budgeting rate, but only 31 % maintain emergency funds. Digital banking usage is at 56 %, with 68 % feeling uncomfortable with digital tools. Formal banking use stands at 43 %, and their average financial knowledge score is 5.4 out of 10 (SD = 1.6). Interestingly, 67 % participate in informal savings groups like samitis or ROSCAs, and 81 % mainly rely on family advice for financial decisions, highlighting strong informal networks. Demographically, 48 % have secondary education, 58 % live in rural areas, and the average age is 34 years (SD = 9.1). The average monthly income is 22,000 BDT (SD = 8600), about the national

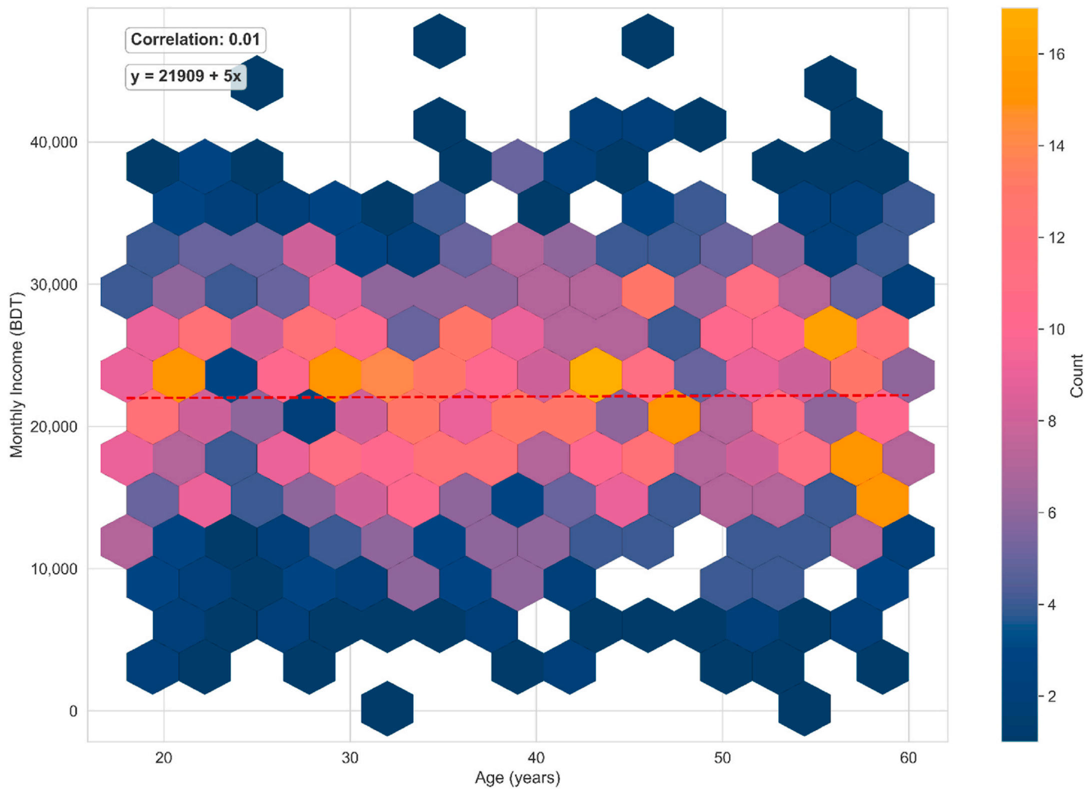

Fig. 8. Age–income distribution of respondents with clustered density.

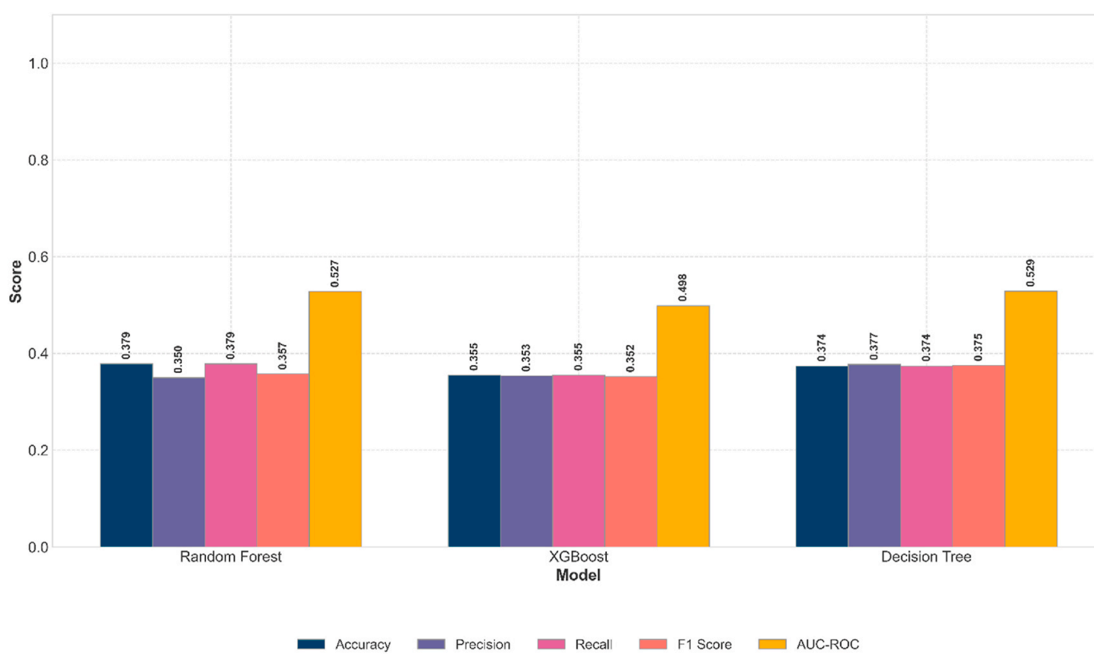

Fig. 9. Model performance metrics across three classifiers.

median. This group demonstrates practical skills through community networks and traditional systems but could benefit from support to access and use formal financial services and digital platforms effectively. It's a transitional group that would thrive with bridge programs linking informal practices to formal services, simple digital tools with good support, and community-based financial education respecting and building on existing strengths.

Cluster 2, known as “Digitally Excluded Traditionalists,” includes 25 % of respondents (n = 267) and faces some significant challenges in various areas of financial capability, as shown in Fig. 14. This group has a savings rate of just 28 %, a mean financial knowledge score of 3.1 out of 10 (SD = 1.8), which is well below the overall average. Their emergency fund maintenance stands at 19 %, digital banking usage is only 15 %, and a large 89 % report feeling only minimally comfortable with digital tools. Ownership of formal banking accounts is at 21 %, with just 12 % using these accounts regularly. Their average trust score in

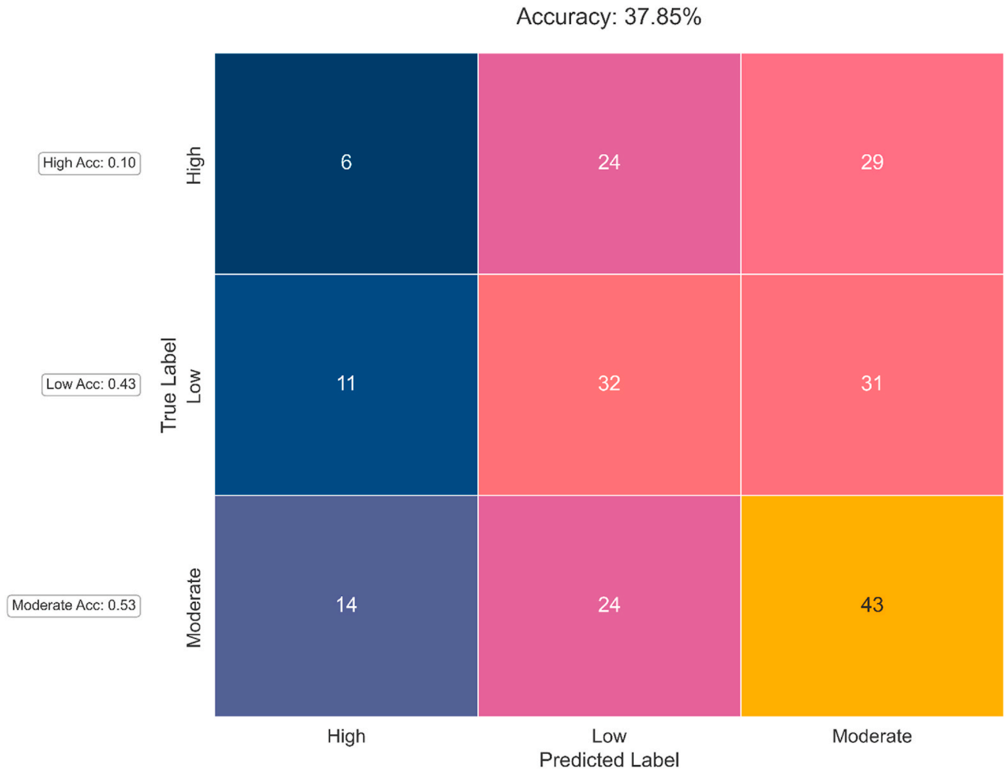

Fig. 10. Misclassification patterns by literacy level.

institutions is 1.9 out of 5, indicating some level of mistrust or skepticism towards formal financial institutions. Despite these hurdles, this group relies heavily on informal financial channels: 73 % borrow from informal lenders, 58 % rely solely on cash transactions, 91 % receive financial support from family during tough times, and 65 % depend on community reciprocity systems. Demographically, this cluster mostly consists of individuals with primary education or less (78 %), many living in rural areas (78 %), with an average age of 42 years (SD = 10.8). Their average monthly income is 15,000 BDT (SD = 6200), which is below the national median. To support this group, targeted interventions that combine basic financial education with digital literacy development are essential. These efforts can help them better integrate into the growing formal financial services and digital payment systems. Yet, it's equally important to recognize and respect their resilience-many rely on strong family bonds and community networks-so these traditional strengths should be incorporated into intervention strategies rather than replaced.

Fig. 15 illustrates a Principal Component Analysis (PCA) visualization of three behavioral clusters within a reduced two-dimensional space, confirming their distinctiveness while indicating some overlap between Clusters 1 and 2, which suggests partial behavioral convergence. The initial two principal components account for 68 % of the total variance: PC1, explaining 42 % of the variance, predominantly reflects digital engagement and integration into formal financial systems; PC2, capturing 26 % of the variance, pertains to financial knowledge and planning sophistication. The pronounced separation of Cluster 0 from the other segments in the upper-right quadrant affirms its classification as a distinct behavioral type with substantially different intervention requirements and market opportunities. The overlap observed between Clusters 1 and 2 in the middle-left region may be attributable to shared educational backgrounds, geographic characteristics, cultural influences, or transitional states wherein individuals evolve between behavioral patterns over time. This visualization offers significant insights for program development, indicating that interventions should consider potential movement between segments and facilitate pathways for progression from lower to higher capability clusters, rather than perceiving segments as permanently fixed categories.

Table 1 presents comprehensive behavioral profiles and intervention recommendations for each of the three clusters, synthesizing demographic characteristics, financial behaviors, digital engagement patterns, and institutional relationships alongside specific intervention strategies, delivery channels, resource allocation guidance, and expected timelines for capability development. The clustering analysis has important implications for financial service providers and policymakers. Cluster 0 represents a market-ready segment that could adopt sophisticated financial products and services with minimal educational support. Cluster 1 represents a bridge segment that requires targeted support to transition from informal to formal financial systems but possesses foundational capabilities that facilitate this transition. Cluster 2 represents a development priority requiring comprehensive support but also significant potential for impact through well-designed interventions.

3.5. Financial pathway analysis and behavioral flow patterns

Sankey diagram analysis revealed multi-step pathways linking location, education, and financial literacy to outcomes like emergency fund ownership and savings frequency. Fig. 16 shows that rural residents have higher emergency fund rates (34 %) than urban residents (28 %), challenging assumptions about rural financial vulnerability. Among highly literate rural residents, 67 % maintain emergency funds versus 52 % of urban counterparts. This suggests rural networks and community-based mechanisms may be more effective for emergency preparedness than urban financial services. The rural advantage likely stems from stronger community support (78 % vs. 54 %), income volatility, cultural emphasis on security, and informal insurance outside formal systems.

Fig. 17 shows pathways from formal education to literacy and savings, confirming high financial literacy often follows graduate education and links with active savings. Among those with graduate education, 58 % have high or moderate literacy versus 42 % with primary education.

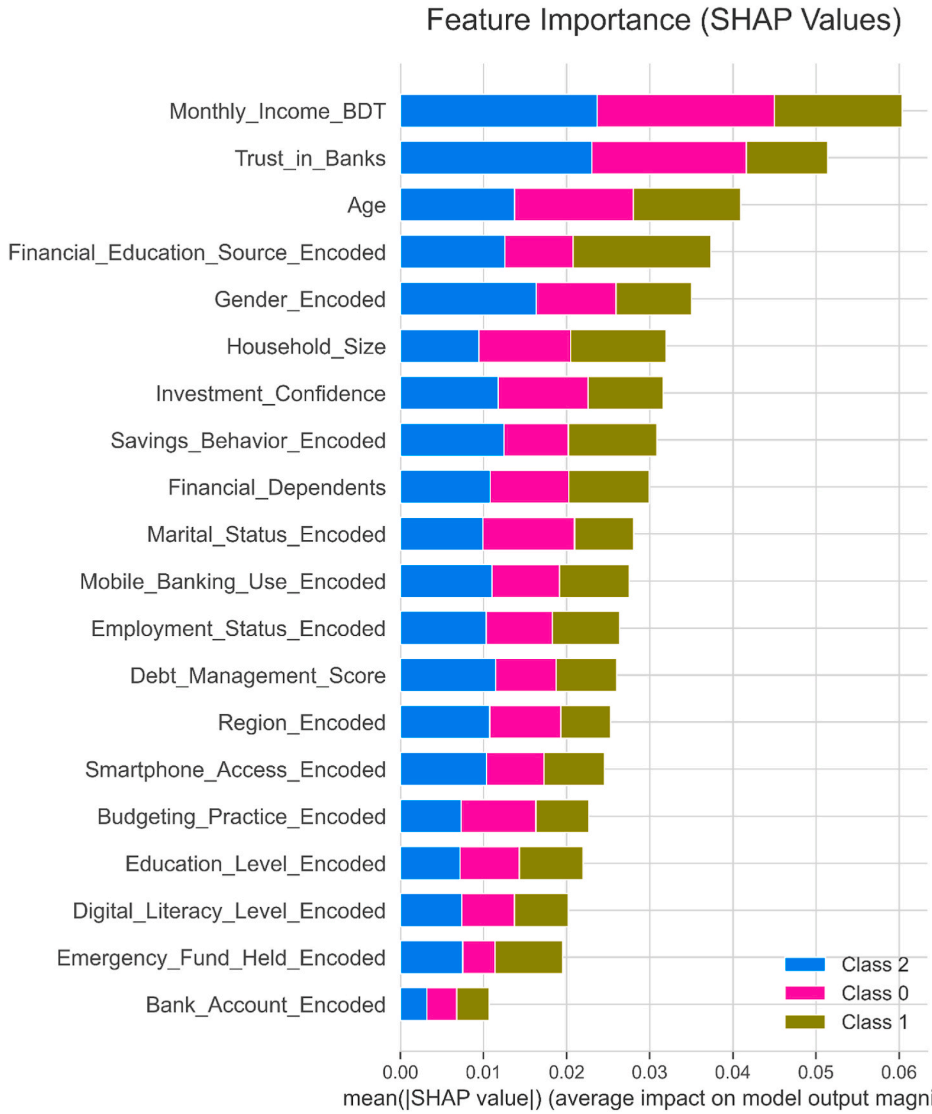

Fig. 11. Top predictors of financial literacy from SHAP analysis.

Also, 72 % of graduates save monthly or more, compared to 48 % with primary education. However, some with limited education develop good savings habits through alternative pathways, with 23 % of primary-educated being high literacy and saving 68 % monthly, similar to 71 % among graduate moderate literacy individuals. These pathways highlight multiple routes to financial capability, suggesting interventions should cater to diverse learning approaches. Figs. 16 and 17 provide frameworks for designing targeted support to boost financial capability. Table 2 summarizes the seven key findings from this analysis, including statistical evidence and implications for policy design and intervention strategies. The table offers an integrated view of how demographic patterns, knowledge-behavior links, predictive modeling insights, and behavioral segmentation work together to inform targeted intervention strategies for Bangladesh's diverse population segments.

4. Discussion

4.1. Reconsidering demographic assumptions in financial literacy

The lack of significant differences across demographic groups challenges assumptions about financial literacy determinants. While international studies emphasize education, urban residence, and male gender as key factors (Lusardi & Messy, 2023; Zaimovic et al., 2023), our findings in Bangladesh show more complex patterns. Rural respondents scored slightly higher, with rural scores 7.3 % above urban, though not statistically significant, and minimal gender disparities suggest informal financial learning plays a bigger role than assumed. Community-based initiatives, informal savings groups, and local financial practices may provide practical education, especially in rural areas with stronger social capital—78 % of rural residents reported reliable support during financial emergencies versus 54 % in urban areas—and seasonal income management needs. These practices include rotating savings and informal insurance. However, since data collection was digital-only, marginalized populations without internet access may be underrepresented, potentially biasing results. The minimal impact of formal education on financial literacy, with ANOVA results showing no significant differences across levels and effect sizes of only 0.2 %, challenges traditional human capital theories (Bayakhmetova et al., 2025; Koskelainen et al., 2023). This aligns with behavioral finance, which emphasizes that knowledge doesn't automatically lead to effective financial behavior, highlighting that interventions focusing solely on knowledge may have limited impact (Sabri et al., 2022). The findings support experiential learning and behavioral reinforcement over classroom-based instruction. Density distributions indicate higher education correlates with more consistent literacy, suggesting formal education influences knowledge application reliability rather than

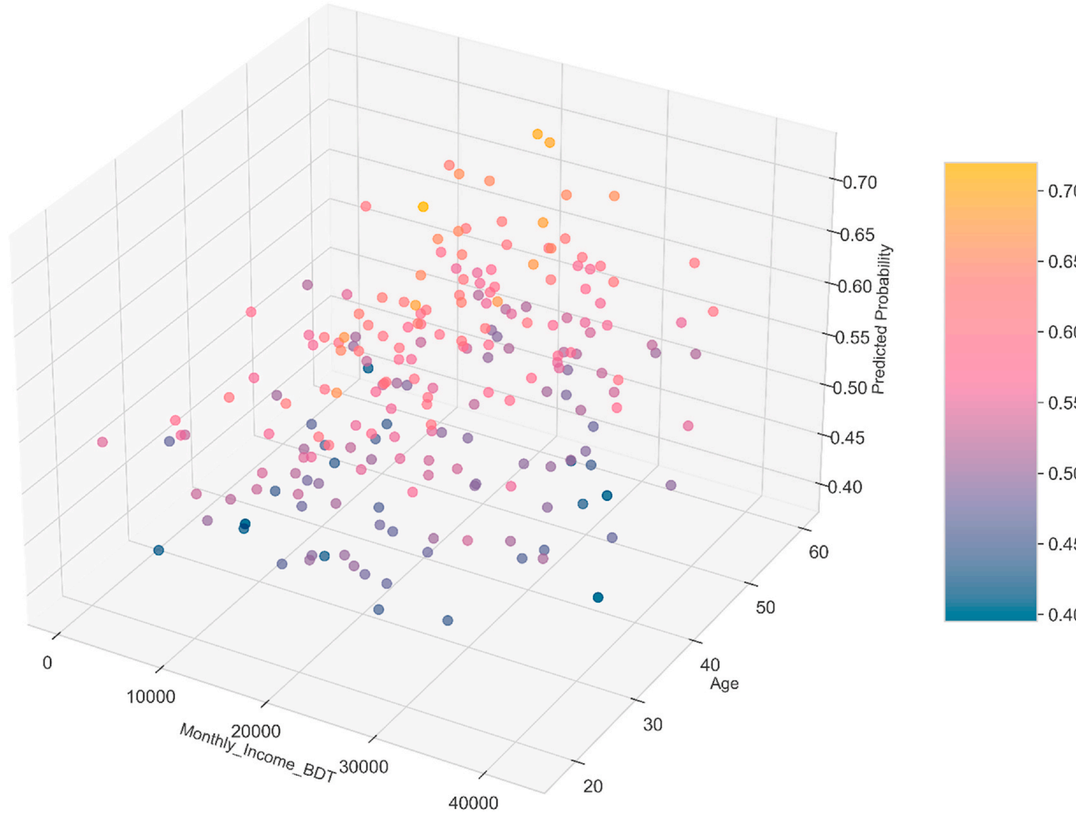

Fig. 12. Interaction effects of age and income on predicted literacy.

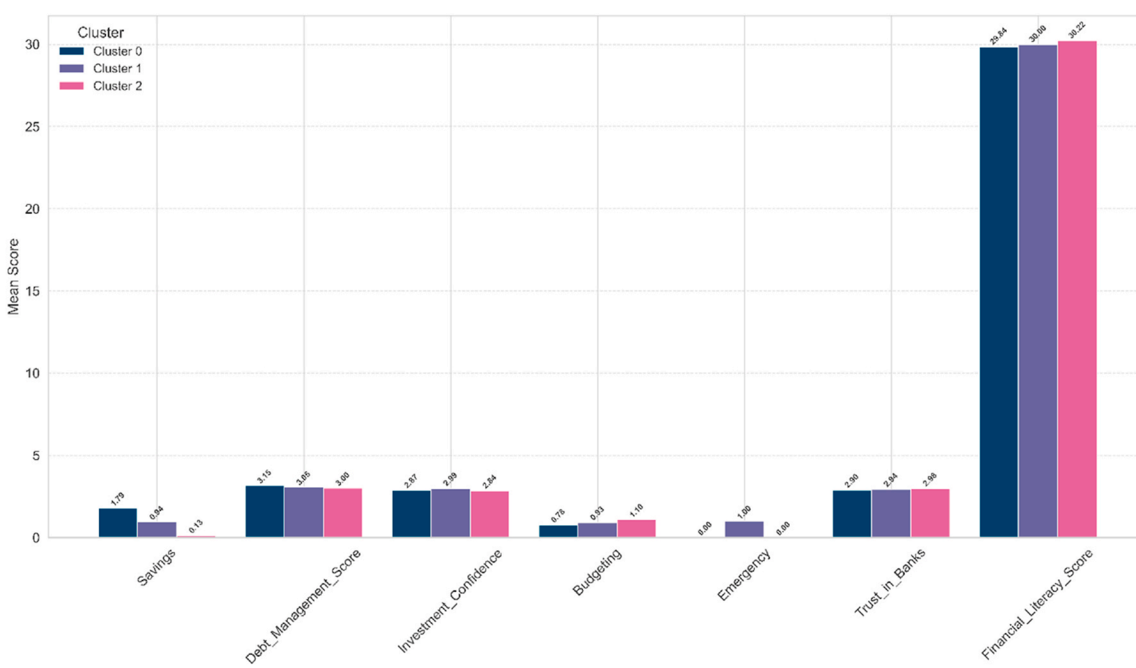

Fig. 13. Financial behavior profiles across respondent clusters.

capability alone. Educational efforts should focus on developing consistent application through practice and feedback, not just knowledge acquisition.

4.2. *The knowledge-behavior gap and behavioral economics insights*

Weak correlations between financial knowledge and behavior, with coefficients below 0.10 and no significance, highlight a key finding with implications for financial education based on behavioral economics. This reflects bounded rationality—people make decisions with limited

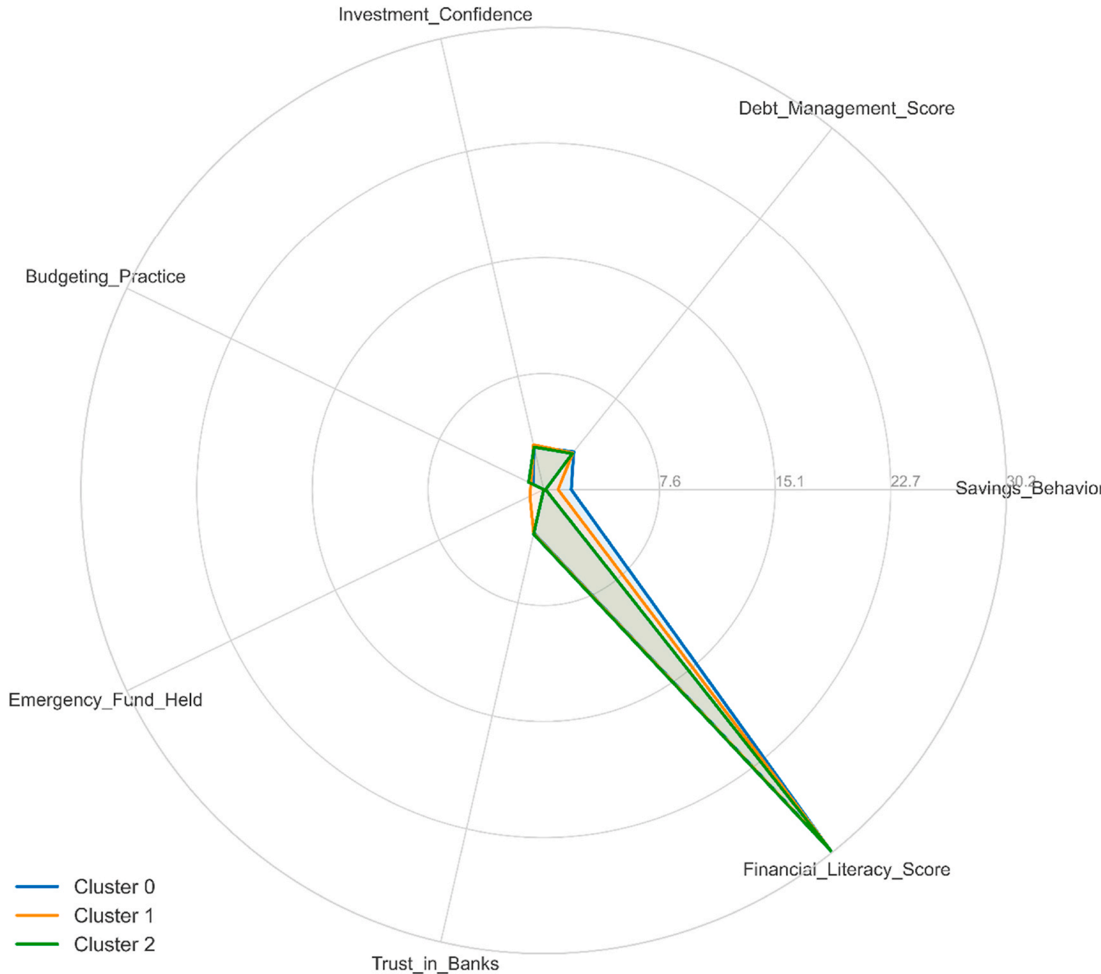

Fig. 14. Behavioral strengths comparison across clusters.

Fig. 15. Principal component view of behavioral clusters.

Table 1
Behavioral profiles of three population clusters identified through k-means clustering with cluster-specific intervention recommendations, delivery mechanisms, and resource allocation guidance. Values represent percentages unless otherwise noted. BDT = Bangladeshi Taka.

Characteristic	Cluster 0: Digitally Literate Planners	Cluster 1: Informally Active but Under skilled	Cluster 2: Digitally Excluded Traditionalists
Sample proportion	34 % (n = 363)	41 % (n = 437)	25 % (n = 267)
Financial knowledge	Mean: 7.8/10 (SD = 1.2)	Mean: 5.4/10 (SD = 1.6)	Mean: 3.1/10 (SD = 1.8)
Savings behavior	87 % regular savers	72 % maintain budgets	28 % regular savers
Emergency preparedness	78 % maintain funds (>3 months)	31 % maintain funds	19 % maintain funds
Digital engagement	• 94 % use mobile banking • 89 % trust digital services	• 56 % use mobile banking • 68 % low digital comfort	• 15 % use mobile banking • 89 % low digital comfort
Formal banking	• 91 % regular users • Trust score: 4.2/5	43 % regular users	21 % have accounts, 12 % use regularly Trust score: 1.9/5
Informal networks	Low reliance	• 67 % in savings groups • 81 % rely on family advice	• 73 % use informal lenders • 91 % rely on family support
Demographics	• 62 % tertiary education • 71 % urban • Mean age: 37 • Income: 35,000 BDT	• 48 % secondary education • 58 % rural • Mean age: 34 • Income: 22,000 BDT	• 71 % primary or less • 78 % rural • Mean age: 42 • Income: 15,000 BDT
Primary needs	• Advanced products • Investment education	• Bridge to formal systems • Emergency preparedness • Digital literacy	• Basic financial literacy • Digital training • Trust building
Intervention type	• Sophisticated financial tools • Robo-advisory services • Advanced investment platforms	• Simplified banking products • Transitional programs • Digital literacy training • Community-based education	• Basic banking onboarding • In-person training • Agent banking models • Trust-building programs
Delivery channels	• Mobile apps • Web platforms • Fintech partnerships	• Community centers • Microfinance meetings • Simplified mobile apps • Cooperative gatherings	• Village meetings • Trusted local agents • Face-to-face support • Agricultural cooperatives
Resource allocation	• 10 % of intervention budget • (minimal support needed)	• 30 % of intervention budget • (transitional support)	• 60 % of intervention budget • (comprehensive support)
Expected timeline	• Immediate engagement with advanced products	• 12–18 months for formal system integration	• 24–36 months for basic capability development
Implementation partners	• Private sector (banks, fintechs) • Investment firms	• NGOs + Private sector partnerships • Microfinance institutions	• Government + NGOs • Community organizations • Agent banking networks
Success metrics	• Product adoption rates • Investment portfolio diversity • Advanced service usage	• Formal account opening • Emergency fund establishment • Digital transaction frequency	• Basic account ownership • Institutional trust improvement • Savings initiation

resources, incomplete info, and time—so even with financial knowledge, they may not apply it consistently (Chawla et al., 2020; Singh et al., 2020). Consequently, interventions should simplify decisions and include support tools, not just increase knowledge. The challenge in translating knowledge into savings behavior, especially due to present bias—overvaluing immediate rewards—shows people understand savings' importance but fail to act. Effective strategies should include commitment devices, mental accounting, and default enrollment to leverage inertia and promote beneficial behaviors (Sultana et al., 2025). Trust in banking was the second-strongest predictor, with a SHAP importance of 0.18, compared to education at 0.09, highlighting financial self-efficacy—confidence in executing financial behaviors. Knowledge raises awareness, but self-efficacy influences success belief (Lusardi & Streeter, 2023). Financial programs should include confidence-building with mastery experiences, peer models, and supportive environments that reduce anxiety and foster skills. Trust in banks and microfinance institutions is crucial, as they serve as literacy educators. Enhancing transparency with clear fees, contracts, and complaint channels could improve literacy more than traditional financial education, which often overlooks trust barriers.

4.3. ML insights and methodological contributions

The modest predictive performance of ML models, with XGBoost achieving an F1-score of 0.52, representing a 58 % improvement over the random classification baseline of 0.33, highlights both the promise and limitations of AI-based financial profiling. While the F1-score may seem modest in absolute terms, it reflects broader challenges in applying predictive models to social data, such as class imbalance—where high-literacy cases make up only 15 %—latent variables not captured in surveys, and measurement noise from self-reported behaviors (Imani et al., 2025). Nonetheless, these models' value extends beyond predictive accuracy to their interpretability and ability to uncover actionable insights. SHAP value analysis provided transparent feature importance rankings, showing that income, digital access, trust in banks, and sources of financial education are the most critical predictors, consistent with growing literature emphasizing that behavioral and structural variables—such as perceived security, usability, and institutional credibility—are just as important as formal education or income levels (Garson, 2021; Singh et al., 2020). The 3D SHAP interaction plots showed that middle-aged individuals (35–50) with moderate income (20,000–30,000 BDT) and digital access had predicted literacy probabilities over 0.65. This group often demonstrates digital adaptability and financial responsibility, making them ideal for fintech education efforts (Jarupunphol et al., 2024). SHAP analysis improves stakeholder trust and makes findings accessible to policymakers, educators, and NGOs for targeted interventions based on transparent insights rather than black-box models. The study shows ML can be responsibly used in social sciences with explainability techniques that ensure transparency and leverage computational strengths for pattern detection (Yang & Xie, 2025).

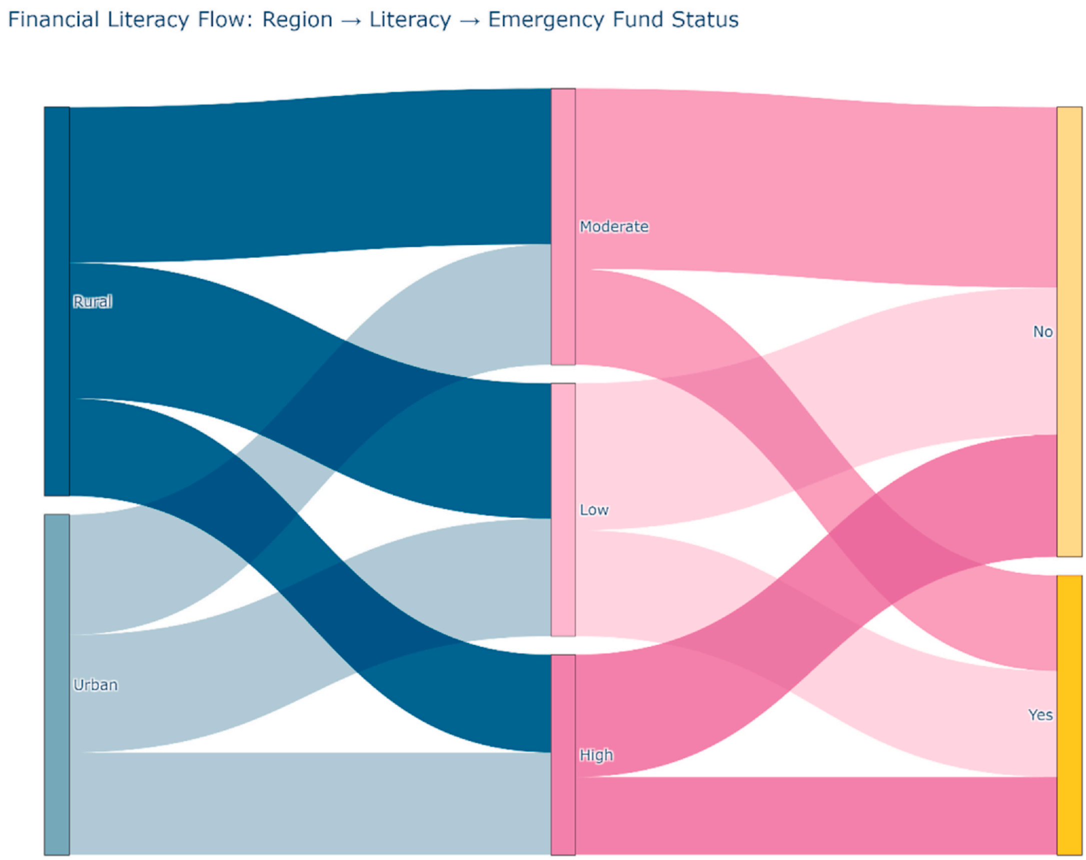

Fig. 16. Pathway from region to literacy to emergency fund ownership.

4.4. Behavioral segmentation within financial capability frameworks

The three-cluster behavioral segmentation identified through unsupervised learning can be meaningfully aligned within established financial capability frameworks while enhancing these frameworks with empirical insights from Bangladesh's context. The World Bank's financial capability approach emphasizes that effective financial behavior results from both ability, which includes knowledge and skills, and opportunity, which involves access to suitable financial services and supportive institutional environments. Our behavioral clusters illustrate this framework: Cluster 0 demonstrating high ability with financial knowledge scores of 7.8 out of 10, combined with high opportunity reflected in 94 % digital banking usage and 89 % institutional trust; Cluster 1 showing moderate ability through informal learning mechanisms, with 72 % maintaining budgets but limited opportunity due to weak formal system integration, with only 43 % using formal banking and 68 % reporting low digital comfort; and Cluster 2 facing both ability constraints, with knowledge scores of just 3.1 out of 10, and opportunity constraints, with 15 % digital banking usage and trust scores of 1.9 out of 5. This operationalization illustrates that capability constraints in Cluster 1 are primarily opportunity deficits rather than ability gaps, indicating that interventions should focus on improving access and building trust rather than solely addressing knowledge gaps (Koskelainen et al., 2023; Zaimovic et al., 2023).

The OECD defines financial literacy as encompassing knowledge, behaviors, and attitudes across key areas, including money management, planning, risk management, and navigating the financial landscape (OECD, 2022). Our behavioral analysis shows that these dimensions do not develop evenly across populations: Cluster 1 exhibits behavioral activity, with 72 % budgeting, but has only moderate knowledge scores of 5.4 out of 10, while Cluster 2 shows low trust, scoring 1.9 out of 5, which limits both knowledge acquisition and behavioral implementation. This uneven development indicates that traditional models assuming simultaneous growth in knowledge, behavior, and attitudes may not fully capture capability development where informal and formal systems coexist. The behavioral segmentation broadens existing frameworks by showing that pathways to financial capability are multiple and nonlinear, rather than progressing uniformly from low to high. Cluster 1, characterized by strong informal capabilities—67 % participating in savings groups—yet weak formal integration, illustrates that individuals can have substantial financial skills through traditional systems, even if they appear to have low capability based on assessments focused on formal financial system interaction (Kamble et al., 2024). This suggests that capability frameworks should explicitly recognize multiple parallel routes to financial well-being instead of assuming a single developmental path from informal to formal system engagement.

4.5. Gender, intersectionality, and inclusive design

Average literacy scores showed minimal gender differences, with men scoring 5.58 and women 5.52, less than 1.1 % apart. However, the distribution of literacy outcomes and interaction effects revealed patterns for intervention. Women benefited more from tertiary education, with an F-statistic of 4.83 and p-value of 0.003, versus men's F of 0.91 and p of 0.44. This suggests women face greater barriers to informal financial learning due to cultural constraints limiting workplace, networking, and social interactions where financial knowledge is shared. Formal education is thus key for women's financial development (Basha et al., 2025; Haag & Brahm, 2025; Khalily, 2008). Interventions should target women's specific challenges, such as limited mobility in conservative communities, caregiving burdens restricting evening or distant programs, and time constraints from household and work duties. Voice-based or vernacular mobile interfaces could enhance inclusivity

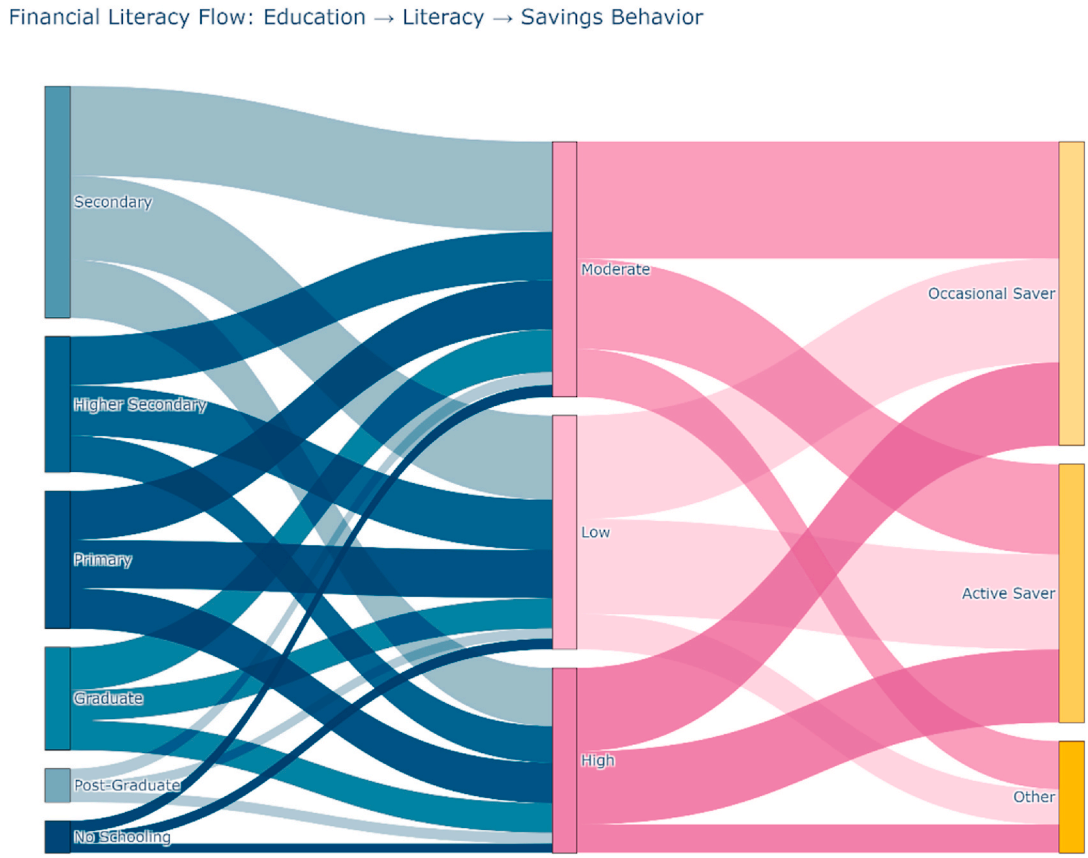

Fig. 17. Flow from education to literacy to saving frequency.

for gender-diverse or low-literacy users facing barriers to text-based digital financial services (Choung et al., 2023; Widyastuti et al., 2024). Inclusive design is vital, improving usability to boost adoption, retention, impact, and market reach for financial providers.

4.6. Digital inclusion and technology-mediated financial behavior

The prominence of digital comfort and institutional trust as key predictors, with combined SHAP importance of 0.29 exceeding education's importance of 0.09, emphasizes that successful financial technology adoption must address both technical and psychological barriers. In Bangladesh, the digital divide extends beyond access to include comfort, confidence, and trust in digital platforms, with 68 % of Cluster 1 and 89 % of Cluster 2 reporting low digital comfort despite some level of digital access. This pattern aligns with research showing that digital financial literacy is a more reliable predictor of good financial behavior than traditional literacy measures alone, indicating that interventions should combine digital skill development with financial education (Choung et al., 2023; Widyastuti et al., 2024). The finding that rural users show higher financial literacy despite lower digital access challenges the assumption that urban living automatically grants digital advantages, suggesting that community-based informal learning mechanisms in rural areas may effectively share digital financial skills through peer networks and social learning even without formal training programs (Clark et al., 2025).

4.7. Implications for policy and practice

Findings from this study highlight several key areas for policy innovation and private sector involvement. First, financial literacy programs should be tailored to behavioral groups identified through segmentation analysis, with evidence-based strategies designed to meet each group's specific needs and abilities. Cluster 2, which makes up 25 % of the population with average knowledge scores of 3.1 out of 10 and trust scores of 1.9 out of 5, would benefit from analog-first training delivered through community partnerships such as village councils and agricultural cooperatives. Meanwhile, Cluster 0, accounting for 34 % with knowledge scores of 7.8 out of 10, could engage in advanced digital investment simulations and robo-advisory services developed by fintech companies (Choung et al., 2023). Resource allocation should align with each segment's needs, with approximately 60 % of intervention resources allocated to Cluster 2's comprehensive support, 30 % to Cluster 1's bridge programs connecting informal and formal systems, and 10 % to Cluster 0's minimal support requirements. This segmentation strategy allows both public and private stakeholders to better allocate resources and maximize impact by focusing on targeted interventions that address specific capability gaps rather than generic programs assuming uniform needs. Trust in financial institutions is the second-strongest predictor after income, highlighting the need for transparent, accessible banking services, especially for marginalized groups. Providers can boost trust by simplifying documentation, clarifying fees, and offering dedicated support, fostering inclusion and loyalty (Lusardi & Streeter, 2023). The correlation between trust and literacy (0.18) shows banks should act as literacy educators, influencing financial capability. Programs targeting trust deficits, like those with low trust scores of 1.9 out of 5, may improve literacy more effectively than purely knowledge-based methods that overlook institutional barriers. Third, mobile internet, smartphone literacy, and app usability are essential for effective financial education, as digital platforms increasingly mediate transactions. Public-private partnerships between network providers, device manufacturers, and financial institutions can offer bundled solutions that improve connectivity, skills, and financial capabilities (Widyastuti et al., 2024; Amit et al., 2024). These models align commercial and development goals, especially with supportive regulations offering incentives like tax

Table 2
Summary of key empirical findings with statistical evidence and policy implications. Abbreviations: F=F-statistic, p = p-value, η^2 = effect size, r = correlation, SHAP = importance, DB = Davies-Bouldin, CH=Calinski-Harabasz.

Finding	Statistical Evidence	Implication for Policy/ Practice
Rural literacy advantage	Rural mean: 5.73 vs Urban: 5.34 F(1,1065) = 2.31, p = 0.13, η^2 = 0.002 7.3 % higher rural scores	Design interventions building on existing rural informal networks (cooperative savings groups, ROSCAs) rather than assuming deficits. Urban programs should emphasize emergency preparedness where rural populations show strength.
Education minimal impact	F(3,1063) = 0.74, p = 0.59, η^2 = 0.002 No significant differences across primary, secondary, higher secondary, graduate	Prioritize experiential learning and community-based education over classroom instruction. Practical application opportunities may be more effective than theoretical knowledge transmission.
Knowledge-behavior disconnect	All correlations $\|r\|<0.10$, p > 0.05 Strongest: r = 0.23 (investment confidence)	Programs must address behavioral barriers (present bias, bounded rationality), confidence building, and institutional access—not just information provision.
Institutional trust paramount	SHAP importance: Trust = 0.18 vs Education = 0.09 2nd strongest predictor after income	Trust-building through transparent fees, responsive service, and simplified documentation is foundational. Banks are literacy educators, not just service providers.
Gender-education interaction	Women: F(3,499) = 4.83, p = 0.003 Men: F(3,562) = 0.91, p = 0.44	Women benefit significantly more from tertiary education. Targeted education programs for women with lower formal education critical for gender equity.
Three behavioral segments	Silhouette = 0.42, DB = 1.08, CH = 287.3 Cluster 0: 34 %, Cluster 1: 41 %, Cluster 2: 25 %	Differentiated interventions based on behavioral typology, not demographics. Resource allocation: 60 % to Cluster 2, 30 % to Cluster 1, 10 % to Cluster 0.
ML modest but meaningful	XGBoost F1 = 0.52, baseline = 0.33 58 % improvement over random	Despite modest absolute performance, models reveal actionable insights through SHAP analysis. Value lies in interpretability and feature importance, not just prediction accuracy.

benefits. Fourth, financial education content should be localized to Bangladesh's context, gender-sensitive, and tailored to address behavioral barriers such as present bias and self-efficacy, not just information gaps (Bhuiyan et al., 2025). Schools, NGOs, and financial institutions should co-develop adaptable materials tested across demographics, combining academic rigor, community reach, and technological capabilities (Haag & Brahm, 2025; Rabeta & Sumi, 2023).

4.8. Limitations and future research directions

Despite its contributions, this study has limitations. Self-reported behavior may suffer from social desirability bias, overestimating positive behaviors like savings and underestimating negative ones like debt, and recall bias, leading to inaccurate retrospective reports (Bertola and Lo Prete, 2025). An online-only survey excludes the most digitally excluded, often the most vulnerable, potentially overestimating financial literacy by missing those without internet access. Cluster 2, at 25 % of respondents but only 15 % digital banking usage, indicates the truly excluded may have even lower financial skills. The cross-sectional design limits causal inferences about whether institutional trust influences literacy or vice versa. Longitudinal studies over 24–36 months could identify triggers for development and causal mechanisms using fixed-effects or difference-in-differences methods. The modest F1-score of 0.52 suggests other variables beyond the survey items could improve the model. Unmeasured psychological traits like financial anxiety, risk tolerance, and locus of control likely influence literacy outcomes (Chawla et al., 2020). Social network factors, such as peer financial behaviors and community norms, may also affect literacy but were not measured. Cultural bias is a limitation; despite adaptation, the survey may reflect Western financial concepts not fully suited to Bangladesh's context, where informal systems prevail. Intersectionality gaps exist because the analysis treated demographic factors independently, failing to consider how multiple marginalized identities, like rural, low-education women, experience compounded disadvantages.

Future research should explore multiple avenues to address these limitations. Incorporating objective behavioral data from mobile money platforms, linking survey responses with actual transaction histories, would validate self-reported behaviors and mitigate measurement error concerns. Longitudinal studies tracking individuals over time would allow for causal inference and insights into literacy development trajectories, including identifying transitions between behavioral clusters. Conducting randomized controlled trials that compare cluster-specific interventions with generic programs would demonstrate the practical benefits of behavioral segmentation for enhancing intervention effectiveness. Cross-cultural validation by applying the framework in other South Asian contexts such as India, Pakistan, Nepal, and Sri Lanka would determine whether patterns are generalizable or specific to certain settings. Adding psychographic measures like personality traits, financial anxiety, and risk tolerance could explore whether psychological factors improve predictive accuracy beyond demographics and behavior. Finally, analyzing algorithmic fairness across demographic groups would assess if ML models produce biased predictions, leading to the development of fairness-aware models that incorporate fairness constraints to prevent unequal error rates across protected groups (Imani et al., 2025; Mustafa et al., 2024).

5. Conclusion

This study analyzed financial literacy among 1067 adults in Bangladesh using ML and behavioral clustering methods, producing findings that challenge traditional views on financial skill development in emerging economies. Three key empirical contributions emerged. First, rural participants slightly outperformed urban residents by 7.3 %, and formal education showed no significant effect on literacy outcomes, suggesting that informal knowledge transfer through cooperative savings groups, agricultural credit associations, and community networks may be more effective than previously acknowledged. Second, ML-based SHAP analysis identified institutional trust (importance value: 0.18), digital comfort, and income as substantially stronger predictors than traditional demographic factors such as education (0.09), while weak correlations between financial knowledge and actual behaviors ($r < 0.10$) challenge models that assume knowledge deficits cause low financial literacy. Third, behavioral clustering revealed three distinct population segments: Digitally Literate Planners (34 %, mean knowledge score: 7.8/10), Informally Active but Underskilled individuals (41 %, score: 5.4/10), and Digitally Excluded Traditionalists (25 %, score: 3.1/10, trust: 1.9/5). These findings advance theoretical understanding in several ways. The prominence of institutional trust over formal education as a predictor extends behavioral finance theory by demonstrating that confidence in financial systems, rather than knowledge acquisition alone, shapes financial capability in contexts where formal and informal systems coexist. The identification of a knowledge-behavior gap supports perspectives emphasizing that financial literacy interventions must address behavioral barriers including present bias, bounded rationality, and self-efficacy rather than simply providing information. Furthermore, the behavioral segmentation approach operationalizes the financial capability framework by showing that ability constraints and opportunity constraints vary independently across population segments, requiring differentiated intervention strategies.

The practical implications for policy and financial service providers are substantial. The behavioral segmentation framework recommends allocating approximately 60 % to comprehensive support for Digitally Excluded Traditionalists, 30 % to transitional bridge programs for the Informally Active segment, and 10 % to advanced product development targeting Digitally Literate Planners. Financial institutions should prioritize trust-building through transparent fees, simplified documentation, and responsive service, recognizing their role as literacy educators rather than solely service providers. The SHAP-based explainable AI framework offers practitioners a replicable approach to translate algorithmic outputs into actionable, segment-specific intervention strategies.

Future research should address current limitations and build upon this work. Long-term studies spanning 24–36 months could help clarify causal relationships in behavior and skill development. Using objective mobile money data would verify self-reports and minimize bias. Randomized controlled trials comparing targeted cluster interventions with generic programs would demonstrate the benefits of segmentation. Cross-cultural validation in South Asian countries such as India, Pakistan, Nepal, and Sri Lanka would evaluate the framework's overall applicability. Future research should also incorporate psychographic variables like financial anxiety, risk tolerance, and locus of control to enhance prediction accuracy. Investigating algorithmic fairness and bias-aware ML models would help prevent inequalities. This study provides a foundation for computational financial literacy by illustrating the potential of ML and behavioral clustering, emphasizing the importance of addressing methodological and cultural challenges.

CRediT authorship contribution statement

Tawhid Ahmed Chowdhury: Writing – review & editing, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation. **Md Ariful Haque Chowdhury:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Md Tahidur Rahman:** Writing – review & editing, Validation, Software, Project administration, Methodology, Funding acquisition, Data curation. **Iftakhar Ahmed:** Writing – review & editing, Validation, Software, Project administration, Investigation, Formal analysis, Data curation. **Nabila Ahmed:** Writing – review & editing, Visualization, Software, Project administration, Investigation, Formal analysis, Data curation. **Md Azizul Islam Tuhin:** Writing – review & editing, Validation, Software, Project administration, Investigation, Data curation. **Abdulla Al Kafy:** Writing – review & editing, Validation, Software, Project administration, Methodology, Formal analysis, Data curation.

Ethics approval statement

This study was approved by the Bangladesh Army International University of Science and Technology Ethics Review Board (Approval Number: 2024-0102; Approval Date: February 2024). All procedures performed in this study involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki Declaration and its later amendments. Digital informed consent was obtained from all individual participants included in the study prior to survey completion. Participation was entirely voluntary with no remuneration offered, and respondents were informed of the study's purpose, data utilization, confidentiality measures, and their right to withdraw at any time.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Copilot and Claude to improve the readability and language of the manuscript. After using these tools/services, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request. The data are not publicly available due to privacy restrictions as they contain information that could compromise the privacy of research participants.

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